

Control-M Role using Ansible

Project Objective:

The objective of this project is to automate the installation and configuration of a **Control-M Agent** using **Ansible**. Since the actual Control-M installer is not available, a **dummy implementation** is created.

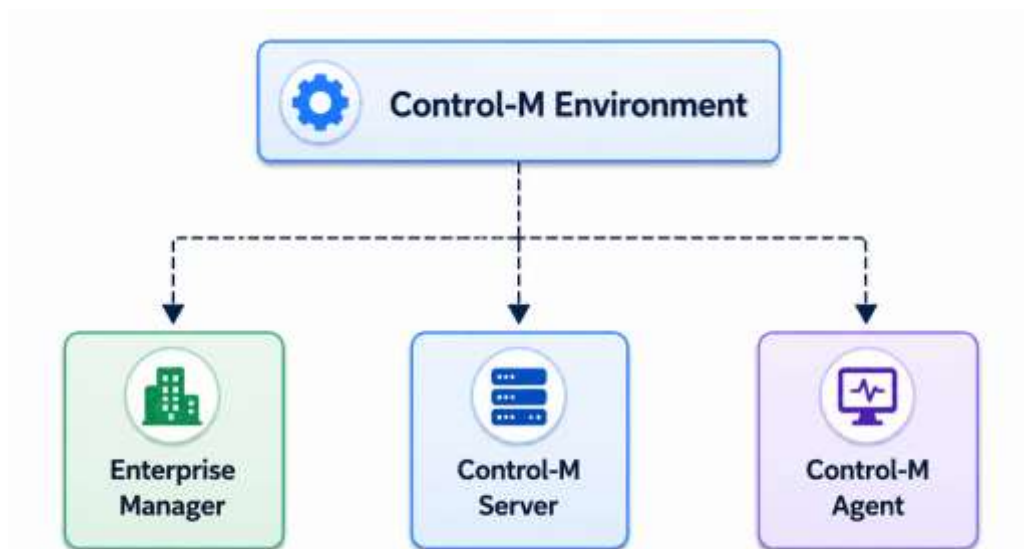
What is Control-M?

Control-M is an **Enterprise Workload Automation and Job Scheduling Tool** developed by **BMC Software**. It replaces repetitive manual tasks with automation.

Typical Tasks:

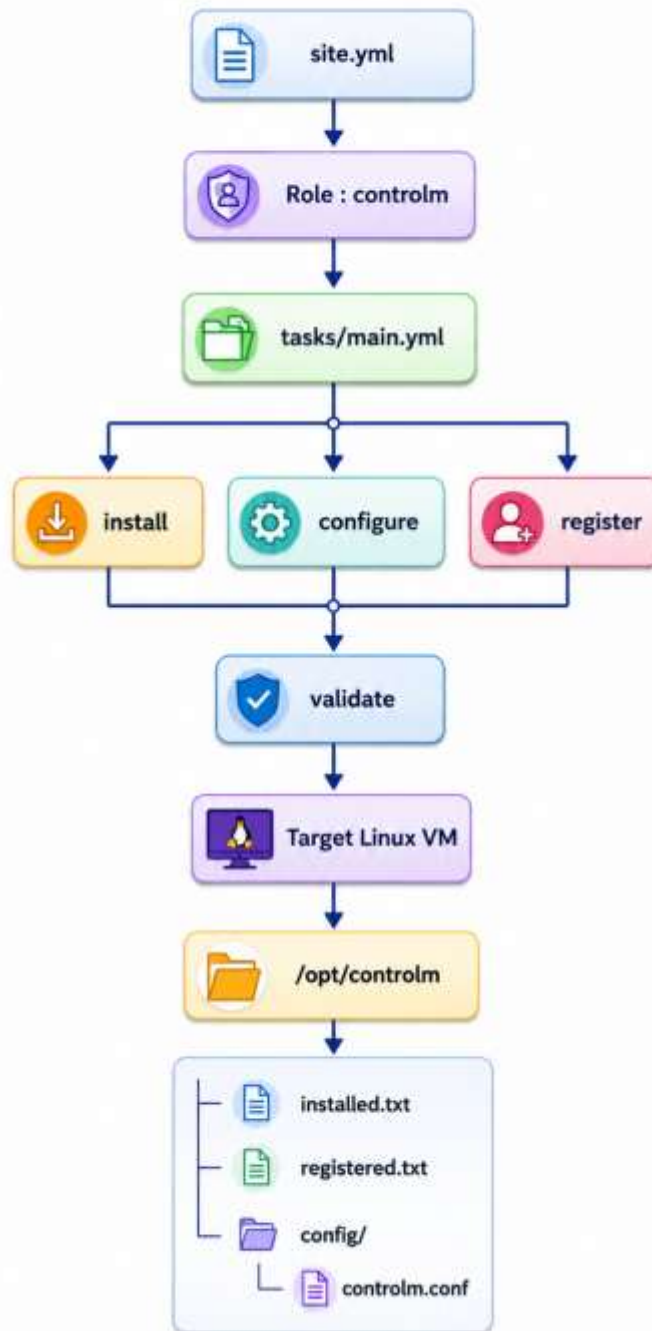
- Database Backup
- ETL Jobs
- File Transfers
- Report Generation
- Batch Processing
- Email Notifications

Components of Control-M



- **Enterprise Manager (EM)** provides a Web Interface used for both Job Monitoring and Job Management.
- **Control-M Server** stores job definitions, schedules jobs, and sends those jobs to the agents.
- **Control-M Agent** is installed on the target server to execute scripts, programs, and batch jobs, and then report their status back to the server.

HIGH LEVEL ARCHITECTURE DIAGRAM



Most commonly used commands:

Command	Purpose / Description
<code>ansible --version</code>	Displays the installed Ansible version.
<code>ansible-galaxy init roles/controlm</code>	Creates the standard directory structure for the Control-M Ansible role.
<code>ansible-playbook playbooks/site.yml --syntax-check</code>	Performs a syntax check on the playbook without executing any tasks.
<code>ansible-playbook playbooks/site.yml</code>	Executes the complete Control-M role on the target VM.
<code>ansible-playbook playbooks/site.yml --check</code>	Performs a dry run to preview changes without applying them.

Deliverables:

```
TASK [controlm : Create registration file] *****
changed: [14.07.20.83]

TASK [controlm : Display registration message] *****
ok: [14.07.20.83] => {
  "msg": "Control-M Agent registered successfully."
}

TASK [controlm : Verify installation marker] *****
ok: [14.07.20.83]

TASK [controlm : Verify registration marker] *****
ok: [14.07.20.83]

TASK [controlm : Display validation result] *****
ok: [14.07.20.83] => {
  "msg": "Control-M validation completed successfully."
}

PLAY RECAP *****
14.07.20.83                : ok=15   changed=4   unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

joy@VirtuallBoc: /Downloads/second-task/project-ansible$
```

```
joylivingstan1@instance-20260806-183204:~$ cd /opt/controlm/
joylivingstan1@instance-20260806-183204:/opt/controlm$ ls
config installed.txt registered.txt
joylivingstan1@instance-20260806-183204:/opt/controlm$ cat installed.txt
joylivingstan1@instance-20260806-183204:/opt/controlm$ cat registered.txt
Registered to dummy-controlm-server
joylivingstan1@instance-20260806-183204:/opt/controlm$
```

Optional Commands:

Command	Purpose / Description
ansible-vault create group_vars/gcp.yml	Creates a new encrypted Ansible Vault file. The user is prompted to set a vault password before creating the file.
ansible-vault edit group_vars/gcp.yml	Opens an existing encrypted vault file for editing. After entering the vault password, the file is decrypted, edited, and automatically re-encrypted when saved.
ansible-vault view group_vars/gcp.yml	Displays the contents of an encrypted vault file in read-only mode after entering the vault password.
ansible-playbook playbooks/site.yml --ask-vault-pass	Executes the playbook and prompts the user for the vault password to decrypt encrypted variables during execution.